CS486 – Design Brief 4
Digital Mockup Prototyping

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1 Prototype Information

Product name: TripFit

Prototype URL: Open the TripFit interactive prototype

Video URL: *To be added after recording.*

Prototype scope: The submitted prototype is a mobile-first digital mockup shown inside a phone frame. It implements two key path scenarios derived from Design Brief 3: a shared group decision in Barcelona and a quick low-effort nearby decision in Lyon. The initial landing interface presents generic needs—"Create a shared trip" and "Quick Low-Effort Nearby"—rather than forcing users to select a persona. In the primary Lina scenario, she creates Barcelona Trip while she is together with her friends, finds nearby friends already using TripFit, and invites them into the group. Lina then adds her own scattered Google Maps pins, saved TikToks, and Instagram reel before returning home while her friends' additions arrive. Once the shared pool reaches 16 ideas, she starts a Group Vibe Check to reach an explainable decision with a backup. The prototype excludes sign-up and registration.

2 Vision Statement and Task Tree

TripFit helps young independent travelers choose what to do next during short city trips by turning scattered saved ideas and nearby options into a small, transparent shortlist that fits their current mood, time, energy, budget, weather, and group situation.

Root goal: Choose a contextually fitting next activity during a short city trip

1. **Start a timely decision:** choose a shared group decision or a quick nearby decision according to the current need.
2. **Gather candidate ideas:** create a shared trip, invite nearby TripFit users, add one's own scattered saves, and return to it once companions' links and pins have arrived, or reuse nearby and previously saved options.
3. **Express current constraints:** select time, location, energy, budget, environment, and social needs through lightweight controls.
4. **Reduce decision load:** filter unsuitable options, present a shortlist, and explain the top fit and remaining trade-offs.
5. **Commit and recover:** align with companions when needed, confirm one option, preserve a backup, and switch routes if the situation changes.

3 Application of Interaction Design Principles

This section explains how the prototype applies the eight interaction design principles required in Design Brief 4.

3.1 Clear visibility of user actions in the UI

TripFit keeps important choices and progress visible as the task proceeds. Lina sees that her five selected ideas have been added before returning home. The Barcelona Trip card then changes from "Collecting group ideas" to "16 ideas collected," while stating that her own ideas are already included. On the context setup screen, selected group energy, budget, environment, time, and location remain visible in their controls. On the decision board, a compact context summary repeats what the recommendation is based on.

Example screens: Add Your Ideas, Home with Barcelona Trip, Set the Group Vibe, Decision Board.

3.2 Design constraints and visual mapping between tasks and controls

The prototype constrains input to lightweight controls: selectable friend and source cards, chips, segmented choices, and a slider. The organizer never needs to type a vague prompt such as "find something nice for us". Instead, each control maps directly to a task: choose which saved sources to bring into the trip, adjust energy with a slider, set budget with dollar-sign chips, choose environment with icons, and set time with predefined chips. Information buttons explain the meaning of these compact settings.

Example screens: Invite Friends, Add Your Ideas, Set the Group Vibe.

3.3 Meaningful feedback to ease evaluation

TripFit gives feedback about both process and output. The invitation screen briefly scans for nearby friends using TripFit, then confirms that four members have joined. After Lina selects her scattered sources, the app confirms that five of her ideas have entered Barcelona Trip. The home trip card shows the remaining contribution progress and changes to a ready state when 16 ideas have arrived. During sorting, a checklist shows that the system is combining ideas, removing options that are too far or too expensive, considering weather, and prioritizing group fit. On the board, a filtered-out explanation and strength/trade-off blocks make the resulting ranking understandable.

Example screens: Group Ready Confirmation, Add Your Ideas, Home with Barcelona Trip, Sorting, Decision Board, Option Detail.

3.4 Breaking a big task into smaller steps

Choosing what to do next during a short trip is cognitively heavy because it involves contribution, constraints, comparison, and agreement. TripFit breaks the group task into manageable stages: create the trip, invite friends, add Lina's own sources, return home as other ideas arrive, review the shared pool, set context, gather sources, sort/filter, compare the shortlist, check group agreement, confirm, and open directions. Once the decision stage begins, a progress stepper keeps the major stages visible: Context, Sources, Decision Board, Group Check, and Confirm.

Example screens: Create Shared Trip, Invite Friends, Add Your Ideas, Home with Barcelona Trip, Set the Group Vibe, Decision Board, Quick Group Check, Confirmation.

3.5 Task closure

The confirmation screen provides explicit task closure. It does not merely navigate to another list; it states "Decision made!" and shows the selected destination, relevant details, and backup status. The primary next action is clear: get directions. This tells the organizer that the group decision is finished and the group can now act.

Example screen: Decision Confirmation.

3.6 Congratulating the user

The final confirmation screen congratulates the user with the phrase "Good choice." This is a small but important confidence cue. The interviews showed that users often hesitate even after narrowing options, so positive feedback helps reduce post-choice anxiety and reinforces that the selected option fits the group's context.

Example screen: Decision Confirmation.

3.7 Clear exit marks

The prototype provides clear exits throughout the flow. Most screens include a back button. The comparison view is designed as a bottom sheet with a handle and downward close button, so it is visually obvious that the user can return to the decision board. The confirmation and directions screens also include clear exits such as back, back to home, and done.

Example screens: Compare Trade-offs, Decision Confirmation, Directions.

3.8 Allowing errors and emergency exits

TripFit communicates recoverability by showing alternate actions such as changing the main choice or skipping a backup. Within the implemented key path, the user saves an alternative and can interactively switch routes by tapping the highlighted alternate destination card on the directions screen. This supports the travel context, where crowding, weather, fatigue, and group preferences can change quickly.

Example screens: Backup Option, Decision Confirmation, Directions.

4 Instructor Review and Iteration

An earlier prototype version was presented using a simpler two-person travel situation based on Tran's context. In that version, users entered their time, location, weather, and preferences, received a suggestion, and adjusted the inputs if the suggestion did not fit. This made contextual recommendation visible, but it did not make TripFit's distinctive purpose clear.

4.1 Feedback received

During the prototype review, the instructor identified the following issues:

- The product needed to focus on a specific persona and a sharply defined pain point instead of covering many kinds of travel decisions.
- The interaction appeared too close to a generic AI assistant: entering context and receiving suggestions did not explain what TripFit does better.
- Showing an obviously poor suggestion, such as an outdoor activity in unsuitable weather, reduced trust in the recommendation process.
- The prototype needed one clear solution to one core problem rather than several loosely related features.

4.2 Design response

The redesign makes Lina's group decision problem the primary, detailed experience. Her problem is not a lack of places to visit; it is that she and her friends already have too many scattered ideas across TikTok links, Instagram saves, Google Maps pins, and messages. The updated prototype therefore begins with Lina creating Barcelona Trip while the friends are together, detecting nearby friends already using TripFit, and inviting them into the group. Lina contributes her own scattered saves first, making her uncertainty and participation visible rather than treating her only as an organizer. She then returns to the landing screen while her friends' additions visibly accumulate in the trip card. When it is ready, she opens the shared pool and starts a Group Vibe Check.

Previous framing: Enter a situation and receive recommendations.
Revised framing: Turn scattered group ideas into one explainable, recoverable decision.

The redesign also makes the recommendation process more accountable. Context is entered through constrained controls rather than a free prompt; the sorting checklist explains the filtering process; the decision board reduces 16 ideas to three realistic fits and states why 13 were filtered out; and option cards show strengths, trade-offs, and risk. Finally, a quick group check, a saved backup, and route switching address the uncertainty of real travel decisions without requiring the user to restart.

Tran remains as a shorter secondary key path scenario, demonstrating that the same principle of reducing options with transparent reasons can support a time-constrained two-person decision. However, Lina's path is the main demonstration because it addresses the more specific pain point revealed through the instructor feedback.

5 Conclusion

TripFit's prototype demonstrates the central design direction developed across the previous design briefs: the product should not generate more travel content, but help users make a confident decision from content and realistic nearby options. Its primary group path turns scattered contributions into an explainable agreement, while its shorter nearby path supports time-constrained decisions. Both conclude with a clear action and a recoverable backup.